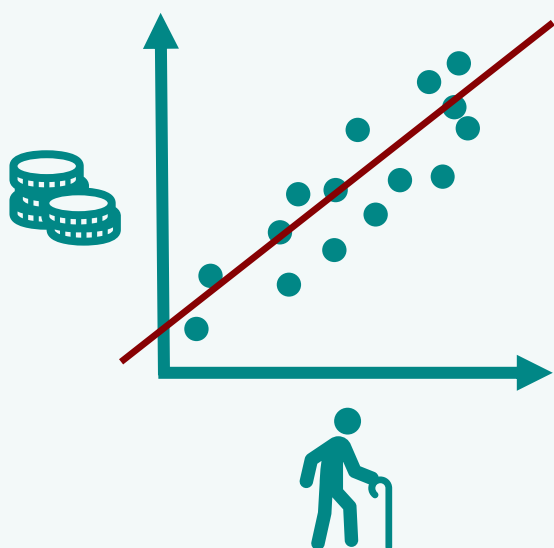


PEARSON CORRELATION

Playbook

Theory & Example



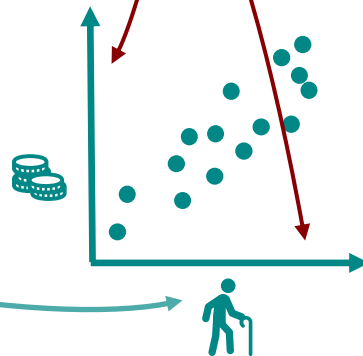


What is the **Pearson correlation**?

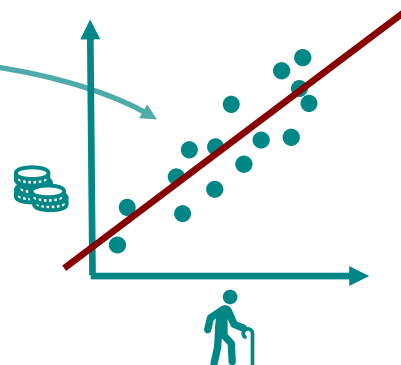
The Pearson correlation analyses the **relationship between two variables**.

For example:

Is there a relationship between a **person's salary** and **age**?



In this **scatter plot**, every single point is a **person**.



If the relationship is confirmed in this example, **salary** can be **predicted** by **age**, using a **regression**.



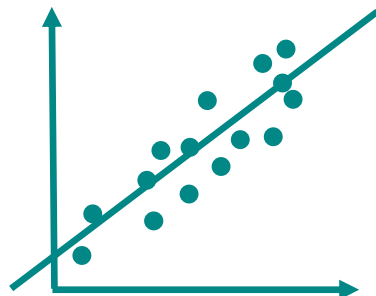
But beware!

It's not always that easy.
There must be a **clear causal relationship** for this.



Just because there is a correlation, you can't tell which way the relationship is going.

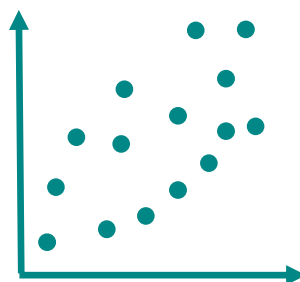
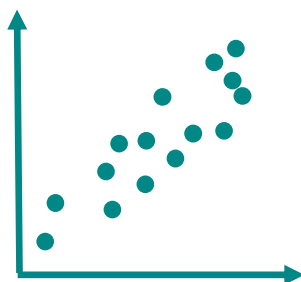
So, with the help of **Pearson correlation**, we can measure the **linear relationship** between two variables.



Here we can determine:

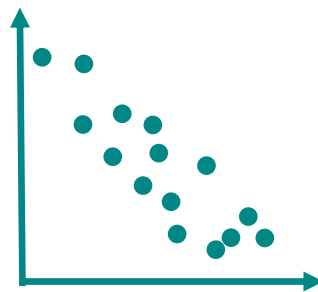
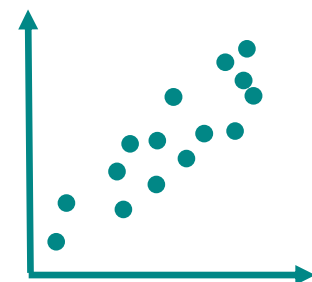
1

How **strong** the correlation is



2

and in **which direction** the correlation goes.



!

We can read both in the **Pearson correlation coefficient r** , which is **between -1 and 1**.



How to read the **strength** of the **correlation**?

The **strength of the correlation**, can be read in a **table**.

Amount of r	Strength of the correlation	
$0.0 < 0.1$	no correlation	If r is between 0 and 0.1 , we speak of no correlation .
$0.1 < 0.3$	low correlation	
$0.3 < 0.5$	medium correlation	
$0.5 < 0.7$	high correlation	
$0.7 < 1$	very high correlation	If r is between 0.7 and 1 , we speak of very strong correlation .

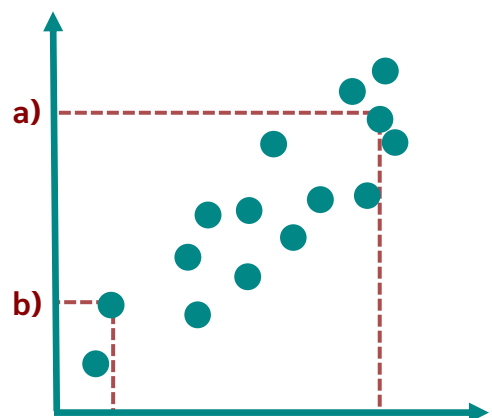
From Kuckartz et al.: Statistik, Eine verständliche Einführung, 2013, p. 213

A

Positive correlation

A positive correlation exists

- when **large values** of **one variable** go along with **large values** of the **other variable**
- or when **small values** of **one variable** go along with **small values** of the **other variable**

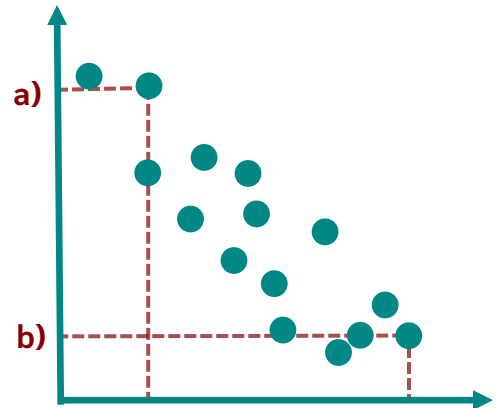


B

Negative correlation

A negative correlation exists

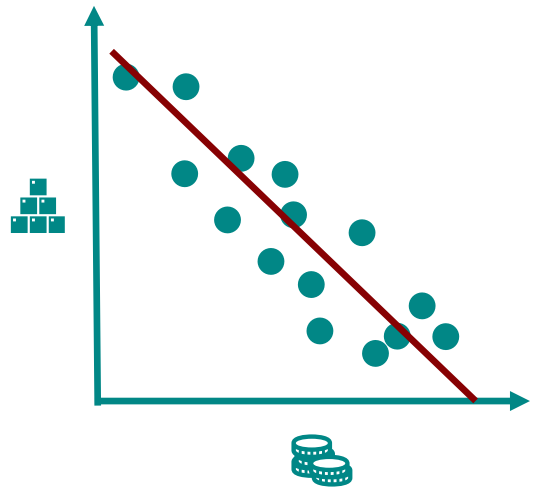
- a) when **large values** of **one variable** go along with **small values** of the **other variable**
- b) or when **small values** of **one variable** go along with **large values** of the **other variable**



A **negative correlation** usually exists between **product price** and **sales volume**.

The result is a **negative correlation coefficient**.

$$r < 0$$

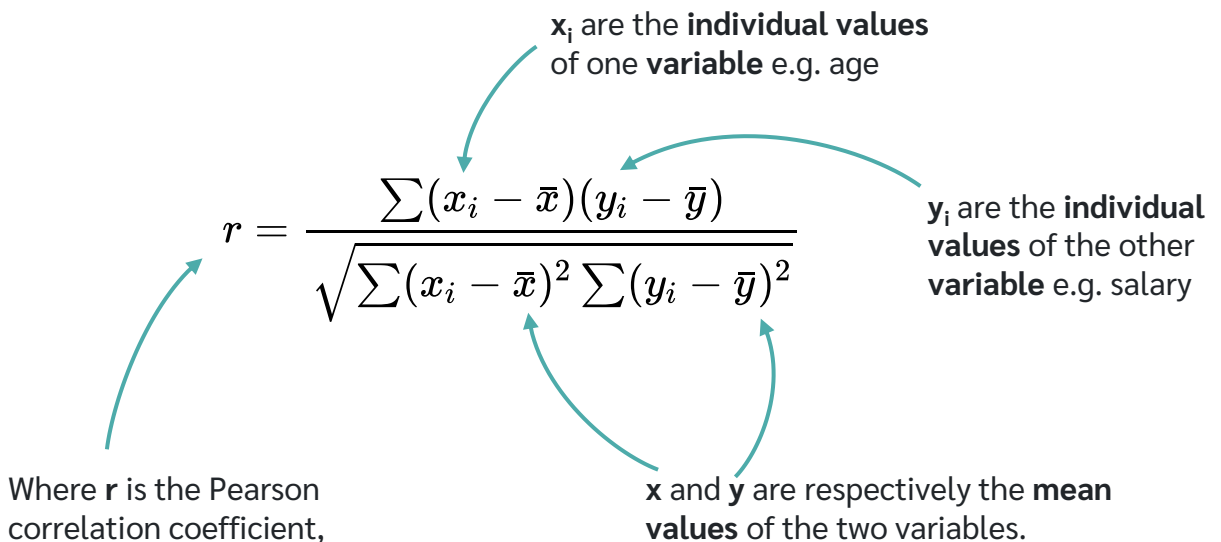




How is the **Pearson correlation** calculated?

The **Pearson correlation coefficient** is obtained via this **equation**:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$



In the **equation**, we can see that the respective **mean value** is first **subtracted** from both variables.



Example:

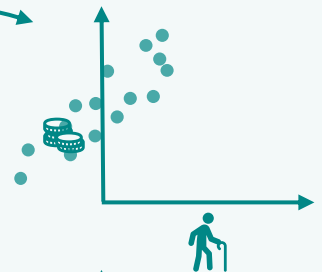
In our example, we calculate the **mean values** of **age** and **salary**.

$$r = \frac{\sum (Age_i - \overline{Age}) \cdot (salary_i - \overline{salary})}{\sqrt{\sum (Age_i - \overline{Age})^2 \cdot \sum (salary_i - \overline{salary})^2}}$$



We then subtract the **mean values** from each person's **age** and **salary**.

$$r = \frac{\sum (Age_i - \overline{Age}) \cdot (salary_i - \overline{salary})}{\sqrt{\sum (Age_i - \overline{Age})^2 \cdot \sum (salary_i - \overline{salary})^2}}$$



$$r = \frac{\sum (Age_i - \overline{Age}) \cdot (salary_i - \overline{salary})}{\sqrt{\sum (Age_i - \overline{Age})^2 \cdot \sum (salary_i - \overline{salary})^2}}$$



We then **multiply** both values.

$$r = \frac{\sum (Age_i - \overline{Age}) \cdot (salary_i - \overline{salary})}{\sqrt{\sum (Age_i - \overline{Age})^2 \cdot \sum (salary_i - \overline{salary})^2}}$$



Then we **sum up** the individual results of the multiplication.

$$r = \frac{\sum (Age_i - \overline{Age}) \cdot (salary_i - \overline{salary})}{\sqrt{\sum (Age_i - \overline{Age})^2 \cdot \sum (salary_i - \overline{salary})^2}}$$

The expression in the **denominator** ensures that the **correlation coefficient** is **scaled** between **-1 and 1**.

$$r = \frac{\sum (Age_i - \overline{Age}) \cdot (salary_i - \overline{salary})}{\sqrt{\sum (Age_i - \overline{Age})^2 \cdot \sum (salary_i - \overline{salary})^2}}$$

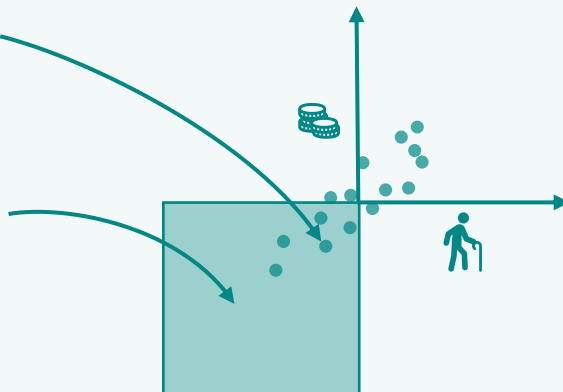
If we now **multiply two positive values**, we get a **positive value**.

So, all values that lie in this area have a **positive influence** on the correlation coefficient.



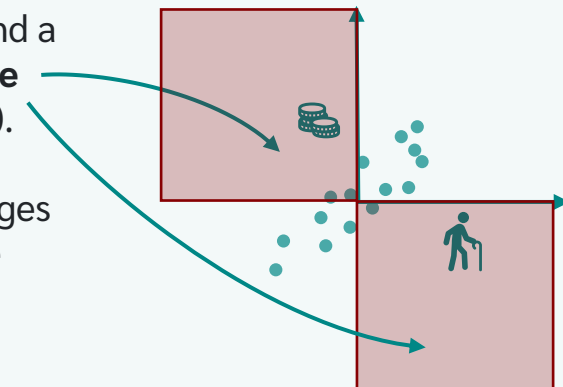
If we now **multiply two negative values**, we get a **positive value**. (Minus times minus is plus).

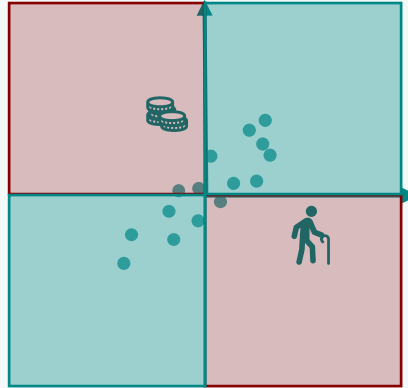
So, all values that lie in this area also have a **positive influence** on the correlation coefficient.



If we **multiply a positive value** and a **negative value**, we get a **negative value**. (Minus times plus is minus).

So, all values that lie in these ranges have a **negative influence** on the correlation coefficient.





Therefore, if our values are predominantly **in the two green areas**, we get a **positive correlation coefficient**.

And thus, a **positive relationship**.

If our values are predominantly **in the two red areas**, we get a **negative correlation coefficient**.

And thus, a **negative relationship**.



If the points are **distributed over all four areas**, the positive terms and the negative terms cancel each other out and we get a **very small or no correlation**.



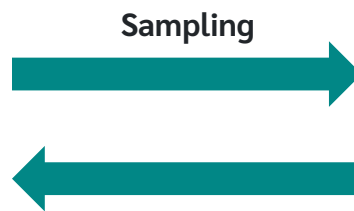
The **correlation coefficient** is usually calculated with data taken from a **sample**.



However, we often want to test a **hypothesis** about the **population**.



Population



Sample

In the case of **correlation analysis**, we then want to know if there is a **correlation** in the **population**.

For this we check, whether the **correlation coefficient** in the **sample** is statistically significantly **different from zero**.



The hypothesis in the Pearson correlation

Null hypothesis

H0

The correlation coefficient **does not differ significantly** from zero.



There **is no** linear relationship.

Attention:

It is always tested whether the null hypothesis is rejected or not!

Alternative hypothesis

H1

The correlation coefficient **differs significantly** from zero.



There **is a** linear relationship.

In our example, we could have the research question:

Is there a **correlation** between **age** and **salary** in the **British population**?



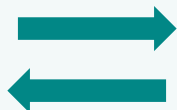
Population



To find out, we draw a **sample** and test whether in this sample the correlation coefficient is **significantly different from zero**.



Population



Sample



The **null hypothesis** then is:

There is **no correlation** between **salary** and **age** in the British population.

And the **alternative hypothesis**:

There is a **correlation** between **salary** and **age** in the British population.

Whether the correlation coefficient is significantly different from zero based on the **sample** collected



can be checked using a **t-test**.

$$t = \frac{r \cdot \sqrt{n - 2}}{\sqrt{1 - r^2}}$$

Where **r** is the correlation coefficient

and **n** is the sample size.

A **p-value** can then be calculated from the **test statistic t**.

If the **p-value** is less than the specified **significance level**, which is **usually 5%**, then the **null hypothesis** is rejected, otherwise it is not.



What about the **assumption** for a **Pearson correlation**?

Here we must **distinguish** whether we just want to calculate the **Pearson correlation coefficient** or whether we want to **test a hypothesis**.



To calculate the Pearson correlation coefficient, only two **metric variables** need to be present.



a person's weight

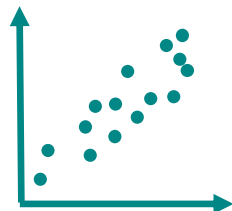


a person's salary

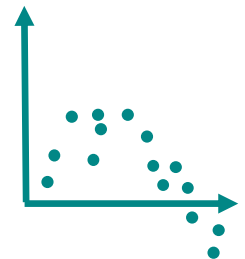


or electricity consumption.

The Pearson correlation coefficient then tells us how large the **linear relationship** is.



If there is a **non-linear relationship**, we **cannot** tell from the Pearson correlation coefficient.



However, if we want to test whether the Pearson correlation coefficient is significantly different from zero, the two variables must also be normally distributed!

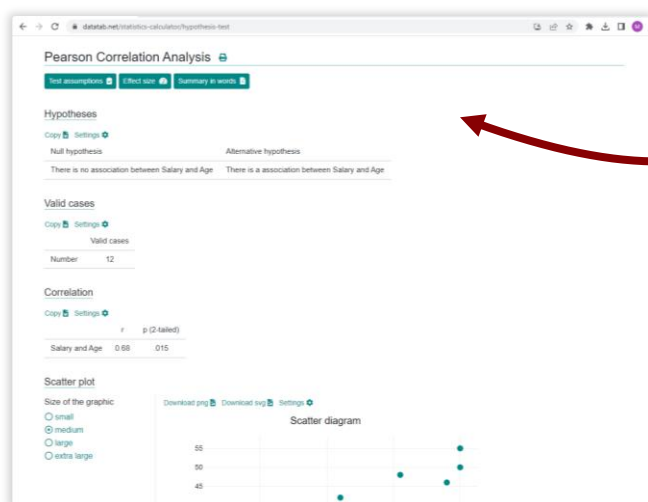
If this is not given, the calculated test statistic t or the p -value cannot be interpreted reliably.



How do we calculate a Pearson correlation with DATAtab?

If you like, you also can calculate the **Pearson correlation** with **DATAtab**

[GO TO DATAtab](#)



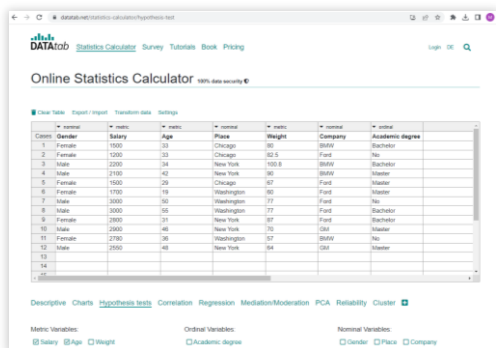
If you like you can of course calculate a correlation analysis online with **DATAtab**.

1

Just copy your data into this [table](#) and click on either the Hypotheses or Correlation tab.

2

If you now click on two metric variables, a pearson correlation will be calculated automatically.



Case	Gender	Salary	Age	Place	Weight	Company	Academic degree
1	Female	1000	33	Chicago	50	Genly	Bachelor
2	Female	1000	33	Chicago	50.5	Ford	No
3	Male	2200	34	New York	102.5	BMW	Bachelor
4	Male	2100	42	New York	90	BMW	Master
5	Female	1500	29	Chicago	57	Ford	Master
6	Female	1700	19	Washington	60	Ford	Master
7	Male	3000	50	Washington	77	Ford	No
8	Male	3000	50	Washington	77	Ford	Bachelor
9	Female	2000	31	New York	57	Ford	Bachelor
10	Male	2900	46	New York	70	Old	Master
11	Female	2700	36	Washington	57	BMW	No
12	Male	3500	40	New York	84	Old	Master
13							
14							

!

If you don't know exactly how to interpret the results, you can also just click on Summary in words!

More about

PEARSON CORRELATION

on our website datatab.net

Feel free
to share!